The force tells us how motion changes

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Net force / ES: fuerza neta / PT: força resultante

Add signed forces along the same axis.

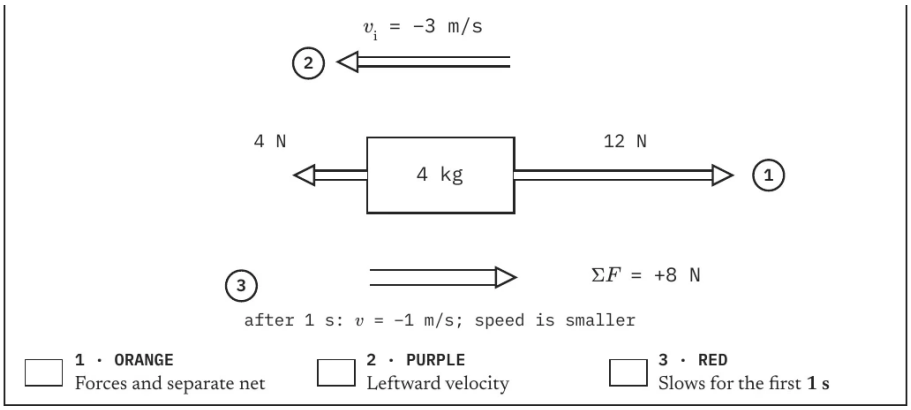
Also on your note page: Acceleration, Velocity, Speed trend.

B READ THE RELATIONSHIP

Add forces first. Then divide by mass.

As you read, find: force · sum · acceleration.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 4 kg cart has forces of 12 N right and 4 N left. Its initial velocity is –3 m/s . Find a and v after 1 s .

Given m = 4 kg F right = 12 N ; F left = 4 N v i = –3 m/s

Rule Sum forces first. Σ F x = ma x . v = v i + a x t .

Work Σ F x = 12 – 4 = +8 N a x = 8/4 = +2 m/s² v = –3 + 2(1)

Answer v = –1 m/s Still leftward; speed is now 1 m/s .

E SAVE FOR THE FINAL MEETING

The direction of XJ's net force tells us its acceleration direction. Its existing velocity must also be known to say whether it speeds up, slows down or turns.

Read the physics: Net force

SUPPORTED READING / read one part, then point to the evidence Name

01 PICTURE IT

Choose one object and a positive direction. Write forces to the right and left with opposite signs. Add before you divide by mass.

02 HOW IT WORKS

A net-force arrow summarizes the individual forces. Do not add that arrow again. Divide the signed total by mass. Acceleration points in the direction of the net force.

03 WHAT TO CHECK

Velocity can point the other way. A left-moving cart with a rightward acceleration slows at first.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 4 kg cart has forces of 12 N right and 4 N left. Its initial velocity is −3 m/s . Find a and v after 1 s . v = −1 m/s Still leftward; speed is now 1 m/s .

Model check: Add forces first. Then divide by mass.

05 CONNECT TO XJ

The direction of XJ's net force tells us its acceleration direction. Its existing velocity must also be known to say whether it speeds up, slows down or turns.

SAY IT BEFORE YOU WRITE

Which two steps give acceleration? I added ___ and ___ to get ___. Then I divided by ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Net force

ES: fuerza neta / PT: força resultante

Meaning in my words	How can you tell?
How would you explain net force to a friend? In my words, this means ___.	Two people push one cart in opposite directions. Why do the arrow directions matter? I can tell because ___.
Example or sketch	**Non-example and why**
One person pushes 6 N right and another 4 N left. What single arrow replaces both? My example shows ___.	A student adds those pushes to get 10 N right. Which push did they count the wrong way? I would change ___ because ___.

Acceleration

ES: aceleración / PT: aceleração

Meaning in my words	How can you tell?
How would you explain acceleration to a friend? In my words, this means ___.	A cart follows a bend at steady speed. Why is it accelerating? I can tell because ___.
Example or sketch	**Non-example and why**
Positive is right. A cart goes from −2 to −6 m/s in 2 s. What acceleration fits? My example shows ___.	A cart travels straight at −6 m/s the whole time. Does the minus sign mean it is accelerating? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Velocity

ES: velocidad / PT: velocidade vetorial

Meaning in my words	How can you tell?
How would you explain velocity to a friend? In my words, this means ___.	Right is positive. What does a velocity of −2 m/s tell you? I can tell because ___.
Example or sketch	**Non-example and why**
How would you show a cart going east at 3 m/s? My example shows ___.	A speedometer reads 3 m/s. What would you still need for velocity? I would change ___ because ___.

Speed trend

ES: cambio en la rapidez / PT: variação da rapidez

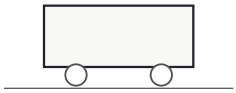
Meaning in my words	How can you tell?
How would you explain speed trend to a friend? In my words, this means ___.	A cart moves left but accelerates right. What happens to its speed before it stops? I can tell because ___.
Example or sketch	**Non-example and why**
Velocities one second apart are −6, −4 and −2 m/s. Describe the trend. My example shows ___.	A student says −2 to −6 m/s means the cart slowed. Which numbers should they compare? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw an object with two opposing forces. Write the signed sum under the net arrow.

Labels you can use: force · sum · acceleration.



Tell a partner

I added ___ and ___ to get ___. Then I divided by ___.

Moving left. Pushed right.

13

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish motion, net force, and acceleration.

CHOOSE ONE JOB TOGETHER

CONTROLS ENGINEER ☐

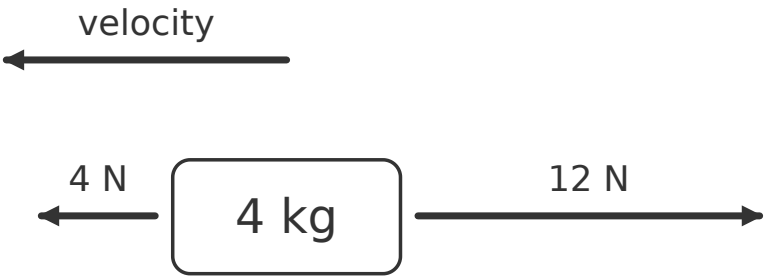
Draw the net-force and acceleration arrows. Tell the operator what happens to speed while the body still moves left.

REPLAY ANALYST ☐

Label separate arrows for motion and acceleration. Explain to viewers how they can point opposite ways.

Word help: net force / acceleration / slowing down

A tracking model moves left. Its **4 kg** body feels **12 N** right and **4 N** left. The status display has separate motion and force arrows.



01 OUR FIRST CALL

What happens to speed at first? Include the net force and acceleration in your call.

It moves ___ while acceleration points ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the updated display. What does a force arrow tell you, and what does it not tell you?

Now we would say ___ because ___.

Switch speakers. What additional information would let you predict when the body stops and reverses?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks